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TASK	INTRODUCTION TO NETWORKING REPORT

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INTRODUCTION TO NETWORKING REPORT

Introduction

In this report, I explored key concepts in networking and cybersecurity, including cryptography, authentication protocols, key exchange methods, and TCP/UDP connections. I focused on understanding how these technologies secure data, verify identities, and enable reliable communication across networks.

Here is a link to my completed module:

https://academy.hackthebox.com/achievement/2380391/34

1.0 Introduction

1.1 Networking Overview

In this section, I learned that networking enables communication between devices using different topologies, media, and protocols. I learned why network segmentation is important for security, as flat networks make it easier for attackers to move laterally and harder to detect malicious activity.

2.0 Networking Structure

2.1 Network Types

In this section, I learned about different network types and how they are categorized based on size, purpose, and connectivity. I learned the differences between WAN, LAN, WLAN, and VPNs, including site-to-site, remote access, and SSL VPNs.

2.2 Networking Topologies

In this section, I learned that a network topology describes how devices are physically or logically arranged and connected in a network. I learned the differences between physical and logical topologies, as well as common types such as star, bus, ring, mesh, tree, hybrid, point-to-point, and daisy chain.

2.3 Proxies

In this section, I learned that a proxy is a device or service that sits between a client and a destination and actively mediates traffic by inspecting its contents. I learned the difference between proxies and gateways, as well as why VPNs are commonly mistaken for proxies. The section also explained the main types of proxies: forward proxies, reverse proxies, and transparent proxies, and how they are used in real-world scenarios such as security filtering, malware defense, penetration testing, and web application protection.

3.0 Networking Workflow

3.1 Networking Models

In this section, I learned about the OSI and TCP/IP networking models and how they are used to describe how data moves across a network. I understood the differences between the two models, their layers, and how data is encapsulated and transferred between systems.

3.3The OSI Model

In this section, I learned about the OSI (Open Systems Interconnection) model and its purpose as a reference framework for understanding how different systems communicate over a network. The OSI model consists of seven layers, each with a specific role, ranging from physical data transmission to application-level interaction. I also learned how data moves through these layers during communication, being encapsulated on the sender's side and unpacked on the receiver's side, ensuring reliable, structured, and efficient data transfer.

3.4 The TCP/IP Model

In this section, I learned about the TCP/IP model, also known as the Internet Protocol Suite, and how it is the foundation of communication on the internet. The model consists of four layers—Application, Transport, Internet, and Link—which combine multiple OSI layers into a simpler structure. I learned that IP is responsible for logical addressing and routing packets to the correct destination, while TCP ensures reliable data transfer and connection control. I also learned how TCP/IP supports applications through ports and relies on DNS for name resolution, allowing users to communicate across different networks regardless of location.

4.0 Addressing

4.1 Network Layer

In this section, I learned that the Network Layer (Layer 3) is responsible for routing data packets from the sender to the receiver across different networks. It handles logical addressing and determines the best path for packets using routing tables and routers. I also learned that protocols like IP, ICMP, and OSPF operate at this layer to ensure packets reach their destination, even across different subnets.

4.2 IPv4 Addresses

In this section, I learned how IP addresses are used to identify devices across networks, unlike MAC addresses which only work within a local network. I learned that IPv4 addresses are 32-bit values split into network and host portions, written in dotted-decimal format, and that subnet masks and CIDR notation are used to define network boundaries and create smaller subnets. I also learned the roles of network, broadcast, and gateway addresses, as well as how binary representation is used to calculate IP addresses and subnet masks.

4.3 Subnetting

In this section, I learned that subnetting is the process of splitting a large IP network into smaller subnets using subnet masks and CIDR notation. I learned how to identify the network address, broadcast address, and usable host range by separating the network and host bits. I also learned basic mental subnetting using powers of two to quickly calculate subnet sizes and ranges.

Question1: Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27

Answer:

The octets: 11111111.11111111.11111111.11100000

128 + 64 + 32 = 224

The decimal representation is thus **255.255.255.224**

Question2: Submit the broadcast address of the following CIDR: 10.200.20.0/27

Answer:

I started by setting all the bits in the host part to 1

There were 5 0s being replaced with 1s so: 1 + 2 + 4 + 8 + 16 = 31

The broadcast address is thus **10.200.20.31**

Question3: Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.

Answer:

/27 split into 4 subnets so I borrowed 2 bits.

New octet address: 255.255.255.248

256 – 248 = 8 so the block size is 8 and therefore each /29 subnet has 8 IPs

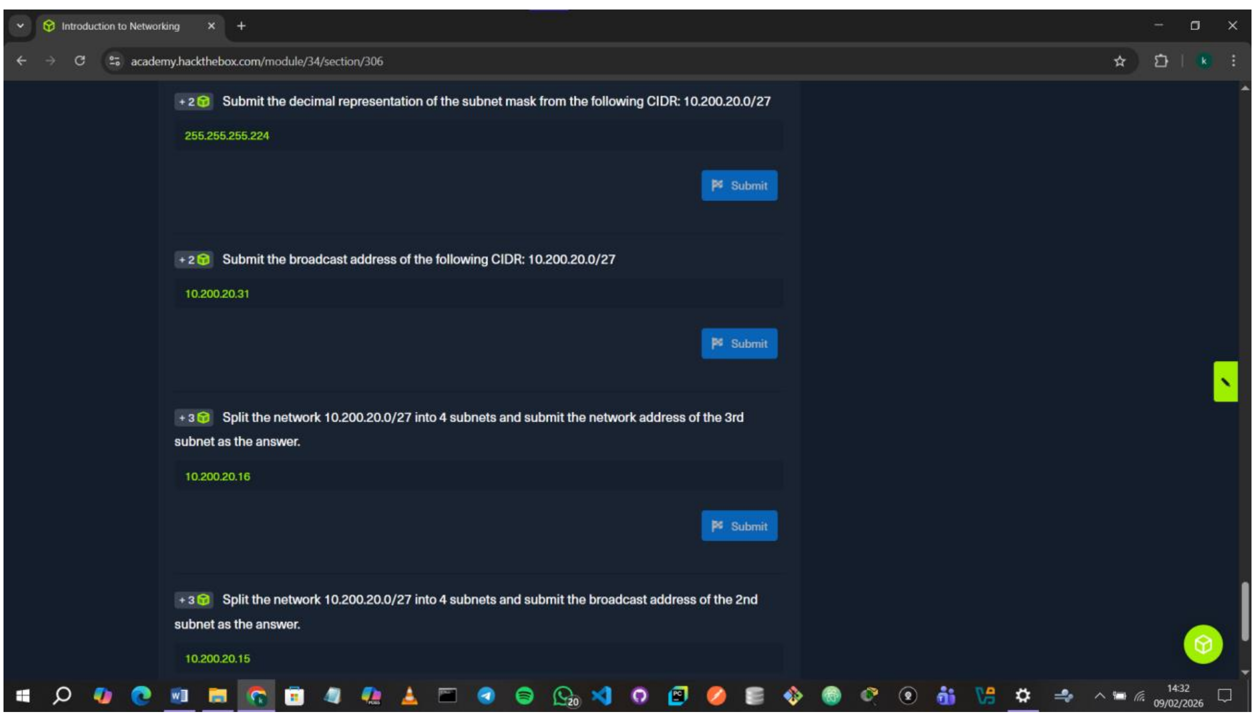
The network address of the 3rd subnet is therefore **10.200.20.16**

Question4: Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer.

Answer:

The ranges are 0-7, 8-15, 16-23, 24-31

The broadcast address of the 2nd subnet is hence **10.200.20.15**



4.4 MAC Addresses

In this section, I learned that a MAC address is a unique 48-bit hardware address used at Layer 2 to identify devices on a local network. I learned how MAC addresses are structured (OUI + NIC), how unicast, multicast, and broadcast work, and how ARP maps IP addresses to MAC addresses. I also learned that MAC addresses can be spoofed and exploited through attacks like MAC flooding and ARP spoofing, so they shouldn't be trusted as the only security measure.

4.5 IPv6 Addresses

In this section, I learned that IPv6 uses 128-bit addresses written in hexadecimal, providing a massive address space and eliminating the need for NAT through true end-to-end connectivity. I learned about IPv6 address types (unicast, anycast, multicast), address shortening rules (::), and how addresses are split into a network prefix and interface identifier, typically using a /64 prefix. I also learned that IPv6 supports features like SLAAC, mandatory IPsec, multiple addresses per interface, and faster, more efficient routing compared to IPv4.

5.0 Protocols & Terminology

5.1 Networking Key Terminology

In this section, I learned the key networking terminology and protocols that form the foundation of IT networking.

5.2 Common Protocols

I learned that devices communicate over networks using standard protocols. **TCP** is reliable and connection-based, while **UDP** is faster but less reliable. Common services like web browsing, email, and file transfer rely on specific protocols and ports. **ICMP** helps with troubleshooting and can reveal information like hops or OS type. I also learned how **VoIP** works and how **SIP** can expose user info.

5.3 Wireless Networks

I learned that wireless networks let devices communicate without cables using radio signals and WAPs. Devices connect via IEEE 802.11, sharing info like SSID, MAC, and supported security methods. I also learned about encryption (WEP, WPA2, WPA3), authentication (LEAP, PEAP), and attacks like disassociation, as well as defenses like hiding the SSID, MAC filtering, and EAP-TLS.

5.4 Virtual Private Networks

I learned that a VPN creates a secure, encrypted connection between a remote device and a private network, letting me access internal resources safely from anywhere. VPNs use clients, servers, encryption, and authentication, with protocols like IPsec and PPTP to protect the data. IPsec encrypts and authenticates traffic, while PPTP creates tunnels but is now considered insecure.

5.5 Vendor Specific Information

I learned that Cisco IOS powers Cisco devices, offering features like IPv6, QoS, security, and support for routing, switching, and VLANs.

6.0 Connection Establishment

6.1 Key Exchange Mechanisms

I learned that key exchange methods let me securely share cryptographic keys to protect communication.

6.2 Authentication Protocols

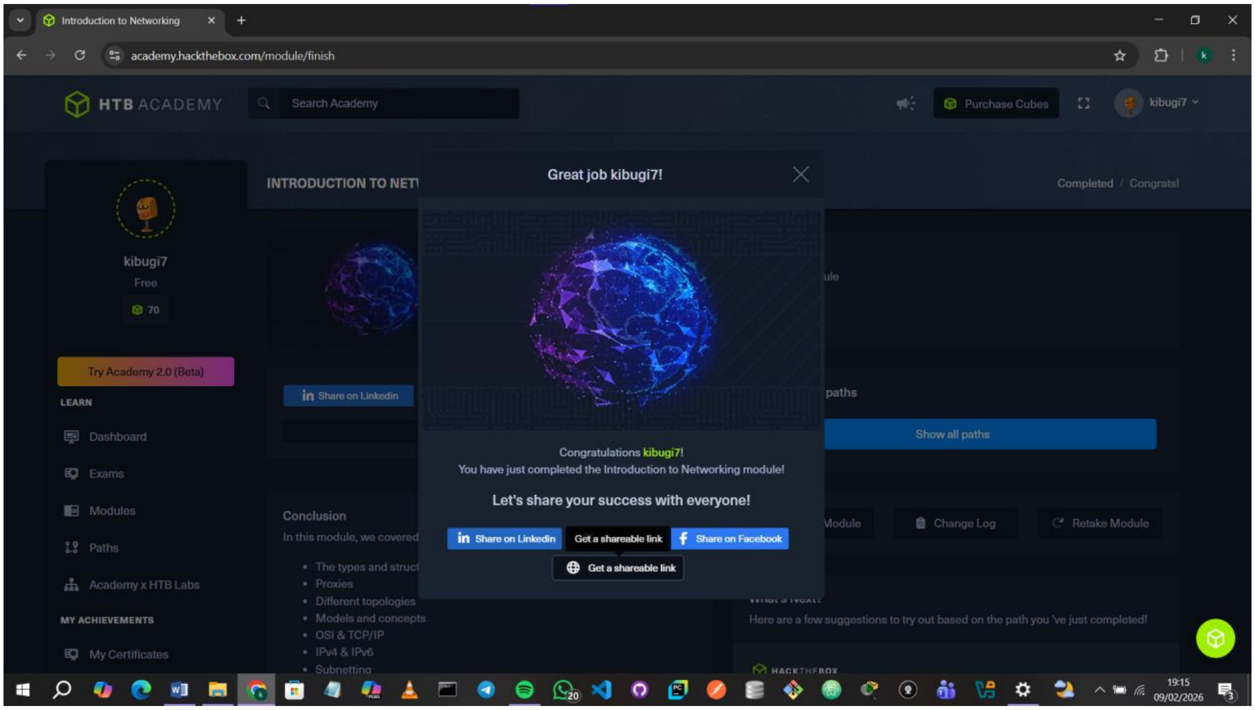
I learned that authentication protocols verify the identity of users and devices in a network, keeping data secure and preventing unauthorized access.

6.3 TCP/UDP Connections

I learned that TCP is reliable and connection-oriented, making it ideal for web pages and emails, while UDP is faster but unreliable, used for streaming and gaming. IP packets carry data with headers for routing and payloads for the actual content. Tools like ping and traceroute help track packet paths. I also saw how TCP/UDP structures work and how attacks like blind spoofing manipulate headers to disrupt connections.

6.4 Cryptography

I learned that encryption keeps data like payments, emails, and personal info safe from unauthorized access. Symmetric encryption uses the same key to encrypt and decrypt data. AES is the most secure today, while DES and 3DES are older standards. Asymmetric encryption uses a public-private key pair, allowing secure communication without sharing keys and enabling digital signatures.



Conclusion

Overall, I gained a deeper understanding of how secure communication works, from encrypting data to establishing trusted connections with TCP, UDP, and authentication protocols like Kerberos or TLS. This knowledge not only strengthens my technical foundation but also equips me to analyze, implement, and maintain secure systems effectively.